



Urban Informatics

Assignment - Week 6

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 16

Total marks: (12 X 1) + (4 X 2) = 20 marks

MCQ/MSQ Question

QUESTION 1: [1 mark]

_____ are a class of deep learning models built using the Transformer architecture and trained on massive text data.

- a. Convolutional Neural Networks
- b. Large Language Models
- c. Clustering Models
- d. Recurrent Neural Networks

Correct Answer: b. Large Language Models

Detailed explanation: Refer to lecture 28, slide 10.

QUESTION 2: [1 mark]

Why are data transformations (e.g., scaling, rotation) applied in machine learning?

- a. To convert datasets into square matrices required by algorithms.
- b. To uniformly redistribute variance across all features.
- c. To make sure patterns becomes clearer and capture maximum variance.
- d. To eliminate feature correlation completely.

Correct Answer: c. To make sure patterns becomes clearer and capture maximum variance.

Detailed explanation: Refer to lecture 27, slide 3.

QUESTION 3: [1 mark]

Which statement best describes a black box model?

- a. A model where internal logic is hidden, focusing only on inputs and outputs
- b. A model built only using qualitative assumptions
- c. A model that cannot be validated or tested
- d. A model with fully transparent parameters and equations

Correct Answer: a. A model where internal logic is hidden, focusing only on inputs and outputs

Detailed explanation: Refer to lecture 28, slide 2.



QUESTION 4: [2 marks]

Match the following Machine learning mechanism with their example-

Learning mechanism	Example task
P. Supervised Learning	i. K-means clustering of land-use zones
Q. Unsupervised Learning	ii. Training an autonomous car through reward signals
R. Semi-Supervised Learning	iii. Predicting house prices from labeled data
S. Reinforcement Learning	iv. Image classification with few labeled and many unlabeled samples

- a. P-i; Q-iii; R-ii; S-iv
- b. P-ii; Q-i; R-iv; S-iii
- c. P-iii; Q-iv; R-ii; S-i
- d. P-iii; Q-i; R-iv; S-ii

Correct Answer: d. P-iii; Q-i; R-iv; S-ii

Detailed explanation: Refer to lecture 28, slide 2.

QUESTION 5: [1 mark]

Which of the following is NOT a supervised learning model?

- a. Support Vector Machine (SVM) used for land-use classification
- b. Random Forest model predicting peak-hour traffic volume
- c. Principal Component Analysis (PCA) applied to urban environmental indicators
- d. Logistic Regression estimating the probability of modal shift

Correct Answer: c. Principal Component Analysis (PCA) applied to urban environmental indicators

Detailed explanation: Refer to lecture 28, slide 5. PCA is an unsupervised learning technique for dimensionality reduction.

QUESTION 6: [1 mark]

SARSA algorithm is a _____ algorithm.

- a. Reinforcement learning
- b. Supervised learning
- c. Unsupervised learning
- d. Clustering

Correct Answer: a. Reinforcement learning

Detailed Solution: Refer to lecture 28, slide 9.



QUESTION 7: [1 mark]

A regression model predicting urban congestion includes three predictors: population density, commercial floor area ratio (FAR), and traffic volume. The model has a high R^2 (0.82), but two predictors show VIF (Variance Inflation Factor) values above 12.

Which of the following statements is MOST accurate?

- a. The dependent variable is poorly measured, which inflates the VIF values.
- b. High VIF indicates that the predictors are strongly related to the dependent variable.
- c. Removing the dependent variable from the model will reduce VIF values.
- d. Even though the model fit is good, the individual regression coefficients may not be reliably interpreted due to multicollinearity among predictors.

Correct Answer: d. Even though the model fit is good, the individual regression coefficients may not be reliably interpreted due to multicollinearity among predictors.

Detailed Solution: Refer to lecture 29, slide 8. VIF measures how much the variance of a coefficient is inflated due to correlations among predictors (not due to the dependent variable). High VIF implies multicollinearity, meaning the model may predict well overall, but it becomes difficult to separate the individual effects of correlated predictors.

QUESTION 8: [2 marks]

In an urban planning study, a researcher wants to predict hourly traffic flow using past time-series traffic data. Which neural network architecture is most suitable for capturing temporal dependencies in such sequential data?

- a. Convolutional Neural Network (CNN)
- b. Long Short-Term Memory (LSTM)
- c. Undercomplete Autoencoder
- d. Deep Feed Forward Neural Network

Correct Answer: b. Long Short-Term Memory (LSTM)

Detailed Solution: Refer to lecture 28, slide 10. RNNs and LSTMs are specifically designed to handle sequential or time-series data. They retain information from previous time steps, making them ideal for modeling temporal dependencies such as hourly traffic patterns in urban planning.

QUESTION 9: [1 mark]

_____ provides a framework for measuring uncertainty (entropy) of a prediction.

- a. Information theory
- b. Optimization theory
- c. Graph theory
- d. Learning theory

Correct Answer: a. Information theory



Detailed Solution: Refer to lecture 27, slide 6.

QUESTION 10: [2 marks]

A city planner is analyzing GPS-based pedestrian movement data in a historic market. The dataset has no predefined labels, contains irregularly shaped clusters, and also includes isolated outlier points caused by occasional tourists. Which clustering method would be most appropriate for identifying the main activity zones while separating the outliers?

- a. K-Means
- b. Hierarchical Clustering
- c. DBSCAN
- d. PCA

Correct Answer: c. DBSCAN

Detailed Solution: Refer to lecture 28, slide 6. DBSCAN, because it can detect clusters of arbitrary shape and identify noise.

QUESTION 11: [1 mark]

When should you use Pearson correlation (r) instead of Spearman rank correlation (ρ)?

- a. For monotonic, non-linear data with outliers
- b. For linear, continuous data without extreme outliers
- c. When strong outliers are present
- d. For categorical variables

Correct Answer: b. For linear, continuous data without extreme outliers

Detailed Solution: Refer to lecture 29, slide 5.

QUESTION 12: [1 mark]

In urban planning research, the Kruskal–Wallis test can be used to-

- a. Compare the mean traffic volumes of two road corridors using normally distributed data
- b. Measure the correlation between land use mix and property prices
- c. Predict traffic congestion levels using multiple independent variables
- d. Compare the median pedestrian satisfaction levels across three different street design types when data is not normally distributed

Correct Answer: d. Compare the median pedestrian satisfaction levels across three different street design types when data is not normally distributed

Detailed Solution: Refer to lecture 30, slide 9.



QUESTION 13: [1 mark]

Which of the following correctly explains why a Train–Test split is essential in a Machine Learning workflow?

- a. To evaluate the model's ability to generalize to unseen data and avoid overfitting
- b. To improve data preprocessing by removing noise from the dataset
- c. To ensure the model perfectly fits the training data
- d. To select the most relevant features for the model

Correct Answer: a. To evaluate the model's ability to generalize to unseen data and avoid overfitting

Detailed Solution: Refer to lecture 26, slide 5.

QUESTION 14: [2 marks]

A city planning department collects large-scale mobility data (GPS traces, land-use mix, traffic counts) to reduce congestion in a historic CBD.

First, A) planners group roads into categories based on traffic volume, land-use intensity, and peak-hour variability. Next, B) they build a Random Forest model to estimate congestion levels for roads in each identified group. Finally, C) they simulate different policy interventions (e.g., parking pricing, restricted vehicle entry, and signal timing changes) to evaluate how congestion levels would change.

Which of the following sequence correctly classifies the following three stages?

- a. A- Prediction; B- Scenario Analysis; C- Clustering
- b. A- Prediction; B- Clustering; C- Scenario Analysis
- c. A- Scenario Analysis; B- Clustering; C- Prediction
- d. A- Clustering; B- Prediction; C- Scenario Analysis

Correct Answer: d. A- Clustering; B- Prediction; C- Scenario Analysis

Detailed Solution: Refer to lecture 26.

QUESTION 15: [1 mark]

Which of the following statements best describes the role of a *pairplot* in multivariate visualization?

- a. It computes exact multicollinearity statistics such as VIF and provides numerical correlation coefficients for hypothesis testing.
- b. It visually displays pairwise relationships among variables, helping identify linear/non-linear patterns, clusters, outliers, and class-wise heterogeneity.
- c. It reduces high-dimensional data into principal components to maximize explained variance.
- d. It automatically selects the best predictive model based on feature interactions.

Correct Answer: b. It visually displays pairwise relationships among variables, helping identify linear/non-linear patterns, clusters, outliers, and class-wise heterogeneity.



Detailed Solution: Refer to lecture 29, slide 10.

QUESTION 16: [1 mark]

Which of the following statements is/are correct about PCA (Principal Component Analysis) and/or LDA (Linear Discriminant Analysis)?

- A. PCA preserves maximum variance
- B. Both PCA and LDA are unsupervised techniques.
- C. LDA is supervised dimensionality reduction technique.
- D. PCA and LDA both maximize variance.

- a. A & B
- b. A & C
- c. C & D
- d. B & D

Correct Answer: b. A & C

Detailed Solution: Refer to lecture 28, slide 5.

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